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Studierichting: Informatica
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$$\int \frac{dx}{4x^2 + 12x + 18} = \int \frac{dx}{(2x = 3)^2 + 9}$$

Stel $2x + 3 = t \Rightarrow 2dx = dt \Rightarrow dx = \dfrac{dt}{2}$

$$= \frac{1}{2} \int \frac{dt}{t^2 + 9} = \frac{1}{2} \int \frac{dt}{t^2 + 3^2}$$

$$= \frac{1}{2} \cdot \frac{1}{3} \operatorname{Bgtan}\left(\frac{t}{3}\right) + C = \frac{1}{6} \operatorname{Bgtan}\left(\frac{2x+3}{3}\right) + C$$

References